Advanced Materials

ADVANCED MATERIALS
www.advmat.de

RESEARCH ARTICLE OPEN ACCESS

Eutectozymes as Soft Hybrid Materials for Advanced Biocatalysis

Manuel Eduardo Martinez Cartagena[1] | Lucia Suarez[1] | Aitor Ontoria[1] | Francisca J. Benitez[2] | Elixabete Rezabal[2] | María Soledad Orellano[1] | Marcelo Calderón[1,3] | Cristián Huck-Iriart[4] | Agustin S. Picco[5] | Matias L. Picchio[1,3] | Ana Beloqui[1,3]

[1]POLYMAT, Applied Chemistry Department, Faculty of Chemistry, University of the Basque Country UPV/EHU, Donostia-San Sebastián, Spain | [2]Donostia International Physics Center (DIPC) and Polymers and Advanced Materials Department, Faculty of Chemistry, University of the Basque Country UPV/EHU, Donostia-San Sebastián, Spain | [3]IKERBASQUE, Basque Foundation for Science, Bilbao, Spain | [4]Experiment Division, ALBA Synchrotron Light Source, Carrer de la Llum 2-26, Cerdanyola del Vallès, Barcelona, Spain | [5]Instituto de Investigaciones Fisicoquímicas, Teóricas y Aplicadas (INIFTA), UNLP - CONICET, La Plata, Argentina

Correspondence: Matias L. Picchio (matiasluis.picchiop@ehu.eus) | Ana Beloqui (ana.beloquie@ehu.eus)

Received: 31 August 2025 | **Revised:** 17 November 2025 | **Accepted:** 12 December 2025

Keywords: biocatalysis | deep eutectic solvents | enzyme immobilization | eutectogels | eutectozymes

ABSTRACT

The development of soft hybrid materials that combine structural integrity, biocompatibility, and catalytic function is a central challenge in advanced biocatalysis. Here, we introduce eutectozymes, enzyme-loaded eutectogels built from natural hydrophobic deep eutectic solvents (HES) and stabilized through a dual supramolecular polymeric network. This unique architecture not only preserves the conformational integrity of enzymes but also creates confined microcavities acting as protective microreactors, thereby enhancing their stability and substrate affinity. Eutectozymes exhibit remarkable resilience under harsh operational conditions, including high temperatures, extreme pH, and organic solvents, while maintaining high catalytic efficiency and reusability. Their versatility is further demonstrated in environmental remediation, achieving over 90% degradation of recalcitrant dyes, and in antimicrobial applications against resistant bacterial strains. By synergistically merging the stabilizing properties of HES with robust gel matrices, eutectozymes establish a transformative platform for heterogeneous biocatalysis, opening new avenues in biotechnology, bioelectronics, and environmental technologies.

1 | Introduction

The utilization of advanced materials as platforms to support functional proteins, such as enzymes, offers multiple benefits, including enhanced enzyme stability against denaturation, increased resistance to inhibitory compounds, and simplified separation and reuse of biocatalysts [1–7]. Nonetheless, the inherent diversity of proteins, their intricate structures and compositions, as well as their multifaceted functions, make the design and the engineering of these platforms challenging. For this reason, current research in enzyme immobilization technologies explores diverse strategies, including the immobilization of enzymes onto the surface of macro- and micromaterials, as well as the entrapment of biomolecules within porous matrices or the encapsulation into nanoscale carriers [8–11]. In this context, soft immobilization platforms, especially those based on gel systems, have garnered significant attention due to their unique combination of structural integrity, flexibility, and biocompatibility [12, 13]. Therefore, traditional hydrogels, known for their water-swollen 3D polymeric networks, exhibit excellent biocompatibility; however, they are limited by their tendency to dehydrate and their mechanical fragility under stress

[12, 14]. Ionogels, formed by integrating ionic liquids into polymer matrices, have emerged as a viable alternative to hydrogels, offering superior thermal stability and enhanced mechanical properties. Still, their use is limited due to concerns about toxicity, elevated synthesis costs, and environmental issues [15, 16]. To address the limitations observed in both hydrogels and ionogels, this work proposes the utilization of eutectogels — stable, mechanically robust soft materials composed of eutectic solvents or deep eutectic solvents (DESs)— [17–19] as a non-toxic and cost-effective biocompatible scaffold for enzyme immobilization [16, 17, 20]. These materials demonstrate remarkable freezing resistance, non-volatility, non-toxicity, and low production costs, overcoming key limitations observed in both hydrogels and ionogels [16, 17, 20]. Hence, eutectogels represent an innovative class of soft materials mainly formed by immobilizing eutectic solvents within silica, polymeric networks, biopolymers, or low-molecular-weight gelator (LMWG)-based frameworks [17, 18, 20–26]. These systems effectively encapsulate eutectic solvents while preserving their beneficial liquid-like properties within stable, mechanically robust gel matrices [17, 18].

DESs are formed through the strategic combination of hydrogen bond acceptors (HBAs), typically quaternary ammonium salts such as choline chloride, with hydrogen bond donors (HBDs), including urea, glycerol, or carboxylic acids. This combination results in eutectic mixtures that exhibit significant melting point depressions compared to the ideal thermodynamic solution [27–29]. The robust hydrogen bonding networks characteristic of DESs can provide a protective microenvironment that preserves the conformational integrity and catalytic functionality of enzymes, even under demanding operational conditions [30–32]. For example, carbonic anhydrase displayed increased thermal stability and even enhanced residual activity when formulated in betaine: glycerol (1: 2 mole ratio) eutectic mixtures, with spectroscopic evidence of conformational compaction due to the hydrogen-bonding of the DES [28]. Indeed, in the last years, the deployment of unconventional solvents such as DES in biocatalytic processes has emerged as a strategy to mitigate fundamental challenges associated with traditional aqueous media, including enzyme denaturation, insufficient substrate solubility for hydrophobic compounds, and compromised catalytic efficiency under industrially relevant conditions [33–35].

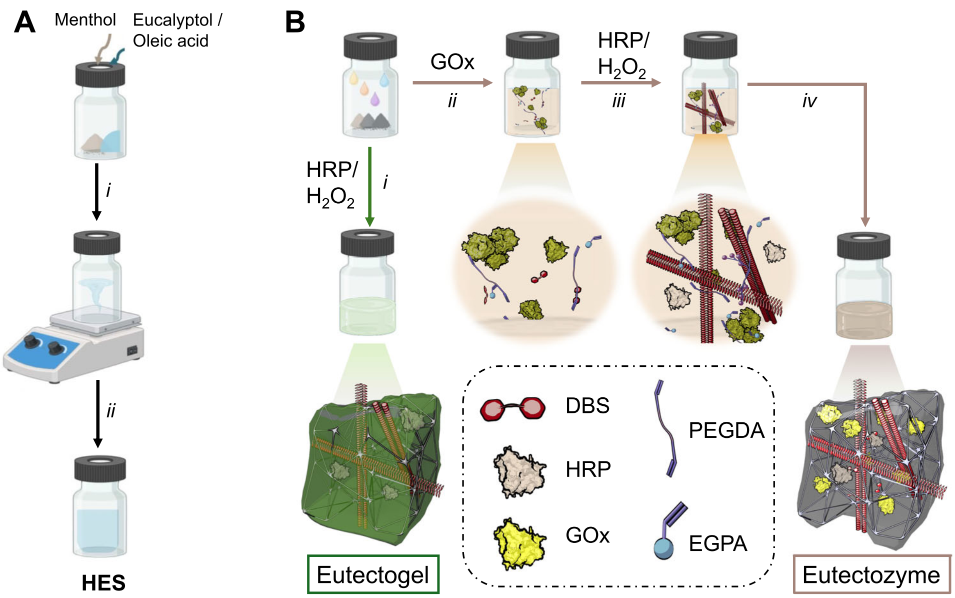
Building upon these advances, we introduced the innovative concept of "eutectozymes", defined as enzyme-loaded eutectogels specifically engineered to enhance enzyme stability, catalytic efficiency, and operational reusability [22]. Eutectozymes respond to a growing demand for biocompatible and sustainable catalytic systems, representing a significant advancement in the design of enzyme-immobilization platforms by synergistically merging the beneficial properties of DESs with the structural advantages inherent to soft gel matrices. This integration creates optimal microenvironments for enzyme immobilization and catalytic function [36]. In line with our original Perspective article [22], where this concept was first proposed and the rationale for employing eutectic solvents within structured materials to improve enzymatic robustness was outlined, the present work further advances its experimental implementation. Compared to existing hybrid biocatalytic systems, such as enzyme organic-inorganic biohybrids—e.g., enzyme-metal organic frameworks or enzyme nanoflowers—where the biocatalyst is also confined within a favorable environment, the gel-like consistency of eutectozymes, combined with their sustainable and biocompatible matrix composition, represents a hallmark of the potential of biocatalytic hybrids for fabricating high-performance materials. Furthermore, compared to existing biocatalytic hybrid systems—such as those based on metal-organic frameworks or traditional immobilization techniques— eutectozymes can be engineered by the use of advanced manufacturing technologies to yield materials with specific sizes and shapes due to their excellent and tunable rheological properties. Moreover, selecting specific DES could impart additional functionalities to the hybrid system, such as conductivity, responsiveness to the external environment, or antimicrobial properties, thereby expanding the landscape of biohybrid materials to the fields of bioelectronics, sensing, or biomedicine [24].

From the wide variety of DESs [37], this study focuses on the utilization of natural hydrophobic eutectic solvents (HESs), composed of water-immiscible components such as terpenes or fatty acids [32], for building enzyme-loaded hydrophobic eutectogels. The resultant eutectogel compositions demonstrate enhanced physical integrity and resistance to solvent leaching in aqueous media, thereby making them particularly advantageous for sustained catalytic performance in aqueous environments and under challenging operational conditions [32]. Particularly, we have optimized the synthesis of double network-stabilized eutectogels based on HES—menthol:eucalyptol (Men:Euc) and menthol:oleic acid (Men:OA)—, a LMWG—1,3:2,4-dibenzylidene-D-sorbitol (DBS)—, and an interpenetrated polymer— poly(ethylene glycol diacrylate-co-ethylene glycol phenyl ether acrylate) (PEGDA-co-EGPEA)— using a one-pot biocatalytic approach. The embedding of glucose oxidase (GOx) into this double-network architecture resulted in the formation of functional eutectozymes. Herein, we have provided an extensive structural and functional characterization of the eutectozyme systems, benchmarking their performance through systematic assessments of mechanical robustness, thermal stability, enzyme stabilization efficacy, catalytic efficiency, and long-term operational durability. Finally, we have demonstrated the practical utility of eutectozymes as reactive materials for removing recalcitrant dyes from aqueous solutions and as powerful antimicrobial materials.

2 | Results and Discussion

2.1 | Synthesis of HES Eutectogels and Eutectozymes

We identified HES as the most suitable mixture for improving the robustness and maintaining the integrity of GOx-loaded eutectozymes in water-based solutions. Hence, a methodology that encompasses, first, the preparation of menthol-based HES solvents (Scheme 1A) and the subsequent fabrication of the soft hybrid materials, namely, eutectogels or eutectozymes, was proposed (Scheme 1B). Men:OA and Men:Euc were prepared at a molar ratio of 2:1 and 1:1, respectively, in accordance with established protocols (see Supporting Information for details) [38].



SCHEME 1 | Schematic representation of the synthesis procedure of the eutectogels and eutectozymes. (A) The initial step involves preparing the HES by vigorously stirring and heating it at 80°C, using mixtures of menthol:eucalyptol or menthol:oleic acid (i), followed by cooling the mixture (ii). (B) For the synthesis of eutectogels, formulations are added to a vial, and the reaction is started by the addition of HRP and H_2O_2. The reaction is maintained for 2 h at 40°C (i). For the eutectozymes, GOx is added to the formulation (ii), and the reaction is initiated upon addition of HRP and H_2O_2 (iii). The mixture is kept for 2 h at 40°C (iv).

TABLE 1 | List of formulations, with corresponding compositions, used in this study.

N°	Code	HES	LMWG	Monomers	Catalyst[a]	Enzyme
1	OA	Men:OA	DBS[b]	PEGDA/EGPEA	HRP/H_2O_2	—
2	OA_C1	Men:OA	DBS	—	—	—
3	OA_C2	Men:OA	—	PEGDA/EGPEA	HRP/H_2O_2	—
4	OA_C3	Men:OA	DBS	PEGDA/EGPEA	—	—
5	Euc	Men:Euc	DBS[b]	PEGDA/EGPEA	HRP/H_2O_2	—
6	Euc_C1	Men:Euc	DBS	—	—	—
7	Euc_C2	Men:Euc	—	PEGDA/EGPEA	HRP/H_2O_2	—
8	Euc_C3	Men:Euc	DBS	PEGDA/EGPEA	—	—
9	OA@GOx	Men:OA	DBS[b]	PEGDA/EGPEA	HRP/H_2O_2	GOx
10	Euc@GOx	Men:Euc	DBS[b]	PEGDA/EGPEA	HRP/H_2O_2	GOx

[a]Catalyst refers to the catalytic system used for the polymerization (biocatalytic system);
[b]DBS is converted to DBSox under these conditions.

It is important to note that, since these HES will ultimately serve as a matrix for embedding GOx, their compatibility with the enzyme was initially verified through an activity assay (Figure S1).

Next, the preparation of the eutectogels based on HES was optimized. Hydrophobic eutectogels are typically obtained through two main methods: (1) the self-assembly of LMWG through non-covalent interactions, mainly hydrogen bonds, creating nanofibrillar networks that entrap the eutectic solvent; [17, 22] or (2) the formation of a crosslinked polymer network in the presence of DES, which provides structural integrity and mechanical stability to the gel [39]. In this study, we initially focused on forming a dynamic, reversible supramolecular network from the self-assembly of a LMWG, namely DBS. However, even though DBS is known to promote the gelation of DES [18], no-gel formation was observed in our HES systems at mild temperatures, i.e., 40°C (Figure S2; Table 1; and Table S1, samples OA_C1 and Euc_C1). Alternatively, a covalently crosslinked network based on the polymerization of hydrophobic EGPEA and PEGDA crosslinker was proposed. This polymerization utilized radical initiation by horseradish peroxidase (HRP) together with hydrogen peroxide (H_2O_2) (Figures S3–S5) [40–42]. Unfortunately, this approach resulted in a discontinuous material that was unable to effectively immobilize the HES (Figure S6; Table 1; and Table S1, samples OA_C2 and Euc_C2).

2.1.1 | Oxidation of DBS During HRP/H_2O_2 Polymerization and Its Role in the Eutectogel Formation

Contrary to initial expectations, DBS, albeit showing a good solubility in our systems, was unable to induce the gelification of HES. Further, the crosslinked polymer failed to convey the required mechanical properties necessary for the eutectozymes. Additional controls, such as OA_C3 and Euc_C3, where the biocatalytic system is omitted, also failed to form gels (Figure S6). Consequently, we decided to repeat the enzymatic reaction in the presence of all the ingredients, i.e., HES, monomers, DBS, and the biocatalytic system (as illustrated in Scheme 1). The reaction was conducted under mild conditions (formulation and conditions collected in Table 1 and Table S1, for samples OA, and Euc), at 40°C for 2 h. Interestingly, a gel with appealing properties was now achieved (Figure S6). This observation led us to investigate the chemical structure of DBS upon polymerization. As shown in the ^{1}H-NMR in Figure 1A, the original signals for the C5-H and C6-H_2 protons (4.75–4.5 ppm) shifted, upon the polymerization, to a downfield singlet corresponding to an aldehyde proton that emerged at δ ~9–10 ppm. Moreover, 2D-NMR, particularly correlation spectroscopy (COSY) and heteronuclear single quantum coherence (HSQC) analysis, provides unequivocal evidence of the oxidation of the sorbitol moiety within DBS (Figures S7,S8). Specifically, the hydroxyl groups of DBS underwent conversion to carbonyls, with the primary hydroxyl group being oxidized to an aldehyde, and the adjacent secondary hydroxyl group transformed into a ketone. No further oxidation to carboxylic acids was detected under these mild enzymatic conditions.

The gel formation properties of oxidized DBS (DBSoxo) in HES were further explored. We could observe that, unlike DBS, isolated DBSoxo yielded a viscous paste with low consistency in the presence of the HES, i.e., Men:OA and Men:Euc mixtures (Figure S9). Therefore, while DBSoxo lacks the 6-OH group, which is involved in the critical H-bonding required for gelation, it continues to form a supramolecular network, most likely due to its retained rigid aromatic-acetal scaffold and newly formed hydrophobic region [43]. In contrast, when the polarity of the solvent was changed using a less hydrophobic eutectic mixture such as menthol: lactic acid (1:1 molar ratio), both DBS and DBSoxo induced gelation (Figure S10 and Table S2). This observation led us to hypothesize that hydrophobic molecules, such as eucalyptol and oleic acid, may disturb the necessary hydrogen bond interactions for the self-assembly of DBS and the resultant gel formation. Therefore, to ascertain the underlying interactions between the gelators and the eutectic components, a preliminary computational analysis focused on the Men:Euc mixture was conducted. Specifically, density functional theory (DFT) calculations were utilized to characterize all conformers of Men:Euc, Men:Euc / DBS, and Men:Euc / DBSoxo within a 3.0 kcal/mol energy window (considering both (−)-menthol and (+)-menthol since the racemic mixture was used for the experiments) (Figures S11–S16 and Tables S3–S5). DFT analysis revealed that the Men:Euc mixture typically interacts through a single hydrogen bond. Interestingly, upon interaction with DBS, this bond is disrupted and replaced by two new hydrogen bonds—one between menthol and DBS, and another between eucalyptol and DBS (Figure 1C, left). Although this interaction is thermodynamically stable (enthalpies of −17.6 and −27.7 kcal/mol), it entails the loss of the Men:Euc internal hydrogen-bond network, thereby disrupting its original structure. This explains the high solubility of DBS observed in the Men:Euc mixture. In contrast, the interaction with DBSoxo does not require breaking the Men:Euc hydrogen bond (Figure 1C, right). Instead, the solvent engages in dispersion interactions with the hydrophobic regions of DBSoxo. This leads to enthalpies of −13.6 and −18.4 kcal/mol, indicating a thermodynamically favorable mixing of DBSoxo with Men:Euc without compromising the internal structure of the solvent.

Consequently, our findings indicate that the implementation of a dual network system —wherein the DBSoxo retains its capacity for self-assembly without disrupting the hydrogen-bonding interactions inherent to the eutectic mixture— facilitates the formation of cohesive eutectogels.

2.1.2 | Formation of Eutectozymes: Enzyme-Embedded Eutectogels

In light of our results on the assembly of HES eutectogels, and considering specific requirements for supported biocatalysis, eutectozymes reinforced with a double-network architecture, i.e., PEGDA-co-EGPEA plus assembled DBSoxo, were further explored. Therefore, the biocatalytic polymerization was repeated, but supplemented with GOx, to produce OA@GOx and Euc@GOx eutectozymes (Table 1 and Table S6). To optimize the matrix composition, which can significantly affect the protein conformation, various parameters were analyzed, including the HES content—ranging from 20% to 60% v/v (Figure S16 and Table S7)— and loaded GOx amount—from 0.6 to 10.3 µg/µL—(Figure S17 and Table S8). A preliminary study suggested that high polymer contents (>50% v/v) led to a decrease in protein activity, most likely due to conformational constraints imposed by the crosslinked network (Figure S18A). Further, a loading of 6.9 µg/µL of GOx was found optimal in terms of catalytic performance (Figure S18B).

2.2 | Chemical and Structural Characterization of Eutectozymes

2.2.1 | HES Composition Affects the Enzyme Distribution Within the Eutectozyme

Several techniques, including X-ray photoelectron spectroscopy (XPS), laser direct infrared imaging (LDIR), fluorescence microscopy, scanning electron microscopy (SEM), wide-angle X-ray Scattering (WAXS), and small-angle X-ray scattering (SAXS), were employed to ascertain the composition and structure of these hybrid materials. XPS was attempted to reveal significant differences in the surface composition between Euc@GOx and OA@GOx. Yet, the obtained results suggest that the protein distribution on the surface of the materials may be similar (Figures S19,S20).

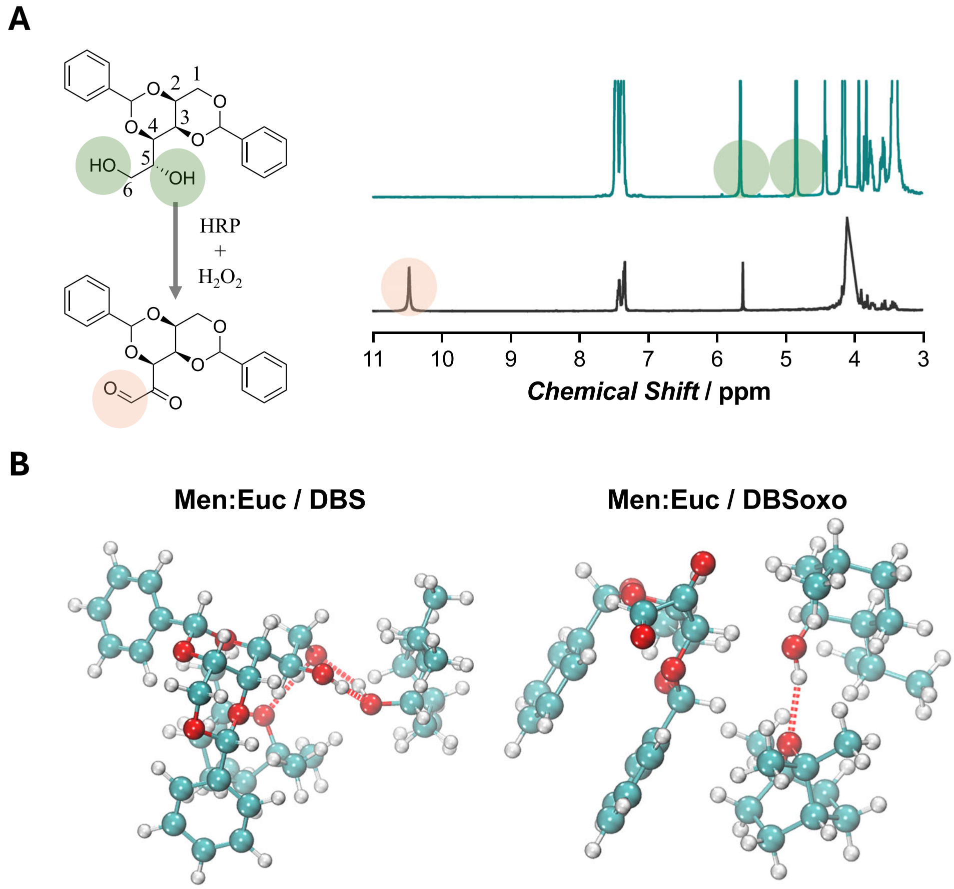


FIGURE 1 | Study of the evolution of DBS during the gelification process. (A) Proposed DBS oxidation reaction and the corresponding 1H-NMR spectra before (green) and after reaction (black), revealing the oxidation of the sorbitol chain within DBS. The disappearance of the protons in positions 5 and 6 of sorbitol (green) and the emergence of a peak corresponding to the aldehyde group (orange) are highlighted. (B) Main conformations for Men:Euc / DBS and Men:Euc / DBSoxo mixtures in gas phase, in the 0–3.0 kcal/mol range. The calculated H-bonds are displayed as red discontinuous lines.

Eutectozymes were subjected to LDIR mapping analysis to determine the homogeneity and spatial distribution of the enzyme within the sample. The localization of GOx enzyme was monitored through the Amide I (~1650 cm^{-1}) band, facilitating the creation of color-scaled concentration maps (Figure 2A,B). Euc@GOx with a loading of 2.8 µg/µL (referred to as Euc@GOx2.8) exhibited an acceptable distribution of the protein across the surface, although some clusters or protein accumulations were observed in the surface wrinkles (as indicated by the red zones and spectrum in Figure 2A,B). The increase of protein concentration to 10.3 µg/µL yielded a more uneven distribution of the enzyme. Similar patterns were noted for OA@GOx (Figure S21). Therefore, higher enzyme loading resulted in the accumulation of enzyme hot spots, leading to an uneven distribution of the protein on the surface of the material.

To gain further insight into the distribution of the enzyme within the eutectozyme, a strategy employing fluorophores was followed. Rhodamine (Rho), a fluorescent compound with favorable dispersion in HES, was incorporated in the synthesis procedure of eutectozymes, giving rise to Euc-Rho@GOx and OA-Rho@GOx formulations (Table S9). Additionally, an alternative strategy using a fluorescent protein was pursued. FTIC-labeled BSA (BSA-FTIC) was added to the reaction mixture instead of GOx, assuming that both proteins would disperse similarly within the HES. Thus, we fabricated Euc@BSA-FITC and OA@BSA-FITC, (Table S9 and Figure S22). With this strategy, both the protein localization and the HES matrix could be monitored by fluorescence microscopy. In particular, the monitoring of Rho unveiled interesting results. As shown in Figure 2C, Rho distribution indicates the presence of microcavities of ca. 50–200 µm, revealed as dark voids in Euc-Rho@GOx. This pattern was similarly presented in OA@GOx (Figure S23A). Plus, the dispersion of Euc@BSA-FITC in spherical clusters shown in Figure 2D suggests that the microcavities were loaded with protein, also in OA@BSA-FITC (Figure S23B,C). These microreactors tend to accumulate at the surface or edges of the material, which explains their functionality (Figure 2E). Hence, the fluorescence analysis suggests the presence of phase-separation domains wherein the protein is localized. We propose that these microstructures (Figure 2F) arise from residual water (16% v/v introduced with the biocatalytic system) or from air droplets immobilized during rapid gelation. This process, driven by DBSoxo self-assembly and EGPEA–PEGDA polymerization, effectively freezes a water-in-HES emulsion and results in the formation of enzyme-confined microreactors within the hydrophobic matrix. With this in mind, we aimed at improving the distribution of these microreactors by applying ultrasound pulses. Interestingly, we observed that

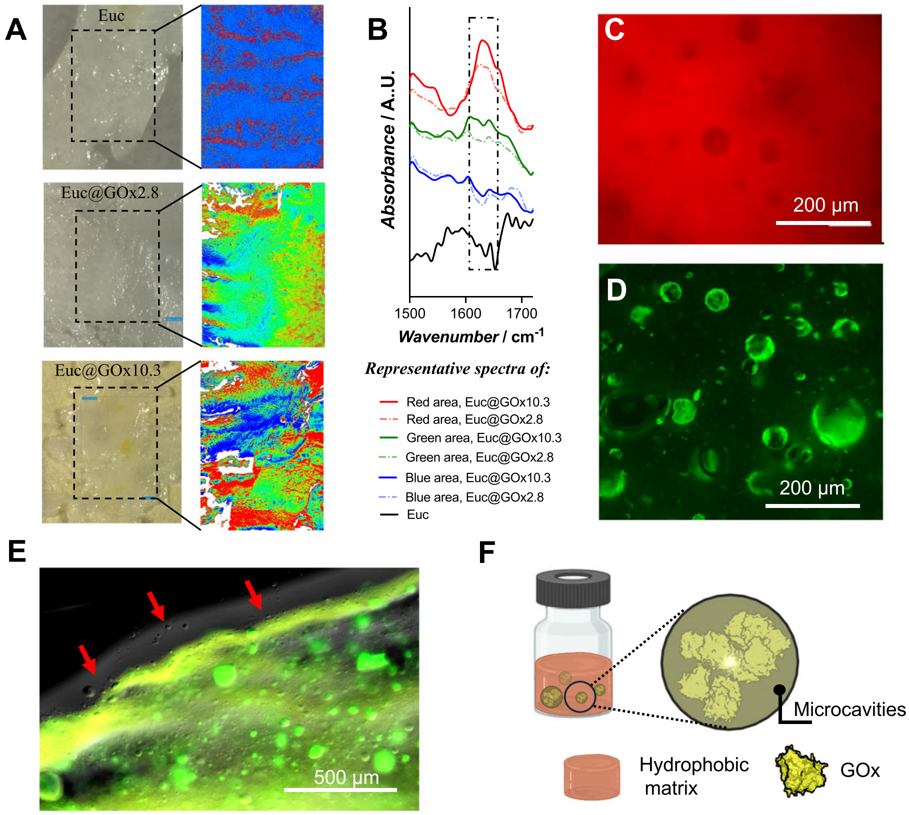


FIGURE 2 | Analysis of the distribution of the proteins within the Euc@GOx. (A) LDIR mapping in which the Amide I band is monitored for Euc, Euc@GOx2.8, and Euc@GOx10.3. The false-color map depicts high/medium/low-intensity Amide I bands in red, green, and blue, respectively. (B) Representative FTIR spectra collected in the blue, green, and red areas from the LDIR mapping shown in (A). Red spectra correspond to the presence of enzyme hot spots. (C) Fluorescence image of the rhodamine B labeled Euc@GOx eutectogel (545–546 nm/excitation and 566–567 nm/emission). (D) Fluorescence image of confined Euc@BSA-FITC within the eutectogel matrix (491–495 nm/excitation and 516–519 nm/emission). (E) Merged fluorescence images of Euc@BSA-FITC loaded with Cy5 fluorophore (649–651 nm/excitation and 667–670 nm/emission). Cy5 was used instead of Rho to improve the visualization of the merged image. Red arrows indicate the accumulation of microcavities on the edge of the material. (F) Depiction that illustrates the hypothesis of the formation of a "frozen emulsion" in which microcavities, filled with protein, are dispersed in the hydrophobic matrix.

the size of the microcavities decreased to <100 μm, and that the sample dispersion improved notably (Figure S24). This finding is particularly significant, as smaller microreactors and improved dispersity have the potential to augment the catalytic performance of the material [44]. However, the application of pulsed ultrasound to GOx-loaded eutectozymes resulted in impaired materials. Therefore, investigating strategies to enhance the dispersibility of microcavities in these hydrophobic materials—such as high-pressure homogenization and microfluidization—while preserving enzyme functionality [45], is of substantial importance and may lead to a considerable improvement in their catalytic performance, opening new opportunities in the design of the heterogeneous biocatalysts.

2.2.2 | Eutectozymes Show Different Levels of Micro- and Nanostructure

SEM was utilized to elucidate the surface morphology of the eutectozymes. As depicted in Figure 3A, Euc@GOx exhibited characteristic features that resemble fibrillar microstructures of ca. 1 μm in thickness. In contrast, the microfibers associated with OA@GOx (Figure S25) were observed to be smoother, displaying a less pronounced network structure. These fibrillar formations are characteristic of gels obtained from the assembly of LMWG [41]. Therefore, DBSoxo offers a well-structured dynamic scaffold for the immobilization of HES and GOx. Moreover, these fibrous structures may serve as a viable template for the formation of polymer networks [46, 47].

Furthermore, the nanostructure of Euc and Euc@GOx (both Euc@GOx2.8 and Euc@GOx10.3) was investigated by SAXS. The scattering curves (Figure 3B) are consistent with a lamellar-like order at the nano/mesoscale [48]. This interpretation is further reinforced by the Kratky plot ($I(q)q^2$ vs. q; inset of Figure 3B), which reveals a peak at ~0.14 nm^{-1}, aligning with a lamellar or long period, $2\pi/q$, of ca. 45 nm. Notably, the incorporation of GOx, even at high concentration, does not significantly disturb the lamellar structure, as evidenced by the remarkably similar scattering profiles observed across all three measured samples. To investigate the lamellar organization of the samples in more detail, the linear correlation function, K(z) [48, 49], was evaluated (Figure 3C). Details on the transformation of SAXS data to K(z) and the subsequent interpretation of K(z) are comprehensively outlined in the corresponding section of the *Supplementary Information*. Analysis of K(z) allowed the

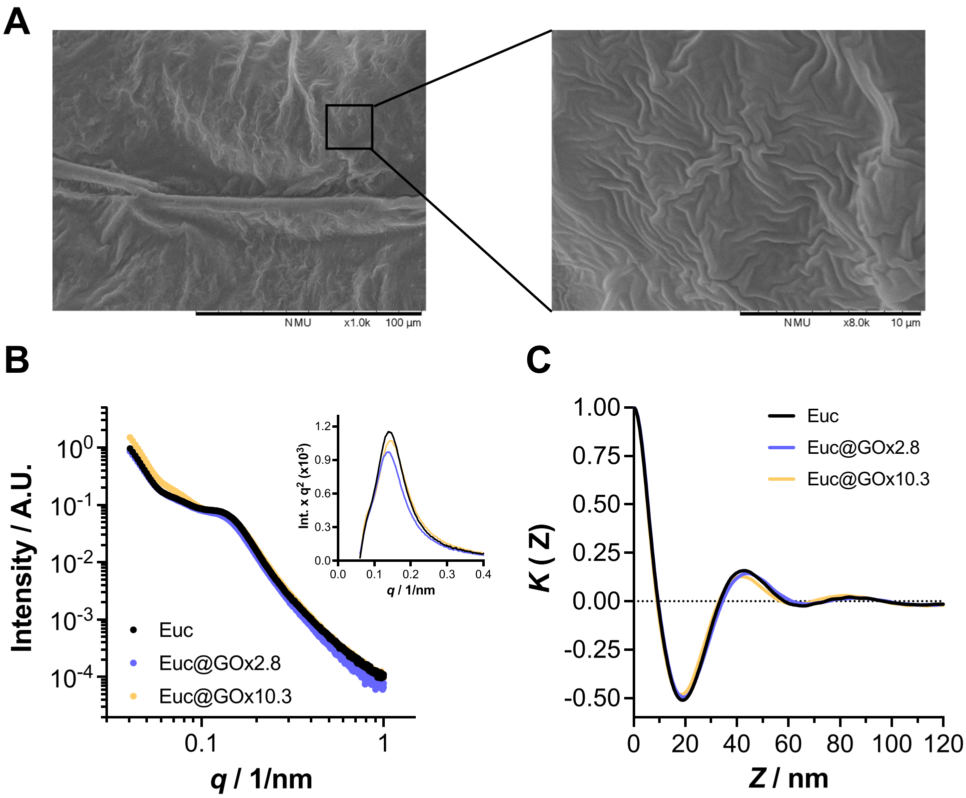


FIGURE 3 | Analysis of the micro- and nanostructures present in Euc@GOx. (A) SEM pictograph that reveals the microfibril structures on the surface of Euc@GOx. The zoomed-in image shows the interpenetration of the fibers. (B) SAXS curves of Euc, Euc@GOx2.8, and Euc@GOx10.3. The inset shows the corresponding Kratky plots. (C) Linear correlation functions *K(Z)* of the three systems derived from SAXS data. The dotted line indicates *K(Z)* = 0.

determination of the characteristic lamellar parameters long period, amorphous thickness, and crystalline thickness in the ranges of 41.3–44.1 nm, 13.7–14.4 nm, and 27.7–29.8 nm, respectively, depending on the formulation (results summarized in Table S10). Across the three eutectogel systems (i.e., Euc, Euc@GOx2.8, and Euc@GOx10.3), the lamellar parameters were similar, indicating that the GOx loads had little influence on the underlying lamellar organization.

The WAXS results at higher scattering angles indicated that all formulations exhibited a similar type of partial crystallization, maintaining a consistent cubic structure. This was evidenced by the progression of Bragg peaks at indices 1, $\sqrt{2}$, $\sqrt{3}$, with a cell parameter of 4.77 nm (Figure S26). However, the intensity of the amorphous contribution (Figure S27) varied among the formulations. The formulations lacking glucose oxidase demonstrated the highest degree of crystallinity (Cr%), 16.6% (Table S10). This suggests that the presence of the protein, when dissolved in the matrix, increases the entropy of the formulation, thereby reducing the overall degree of crystallinity.

Therefore, it seems that employed formulations facilitate the assembly of lamellar-structured gels, wherein crystalline and supramolecular phases are well organized at the nanoscale. Additionally, the presence of microcavities at the macroscale presents a viable opportunity for the confinement of macromolecules. This intricate architecture has been achieved through the formation of a double network based on supramolecular fibrillar motifs assembled from DBSoxo, and a polymer network that likely imparts crystallinity to the material.

2.2.3 | Eutectozymes as Materials With Viscoelastic Behavior

Rheological measurements showed that both Euc and OA rapidly formed gels, but with markedly different kinetics. The complex viscosity (η^*) reached a plateau in ≈10 min for Euc@GOx, whereas OA@GOx required more than 20 min to establish a stable network (Figure 4A). In the solid state, clear differences in the elastic modulus (G′) plateau were observed, with Euc@GOx exhibiting higher stiffness (G′ ≈ 100 Pa), whereas OA@GOx remained a softer gel (G′ ≈ 20–30 Pa) (Figure 4B). This viscoelastic behavior is consistent with the denser interpenetrated fiber architecture revealed by SEM for Euc@GOx [50]. Amplitude sweeps revealed that the yield point of the eutectozymes strongly depended on the HES concentration (Figure S28). Low-HES gels (33 wt%) failed at relatively small deformations (yield point below ∼1%) [51, 52], while high-HES gels (66 wt.%) tolerated much larger strains, sustaining deformation beyond 4% [52]. Intermediate formulations displayed a balanced viscoelastic response, combining moderate elasticity with improved deformability [50, 53–55].

Frequency sweep analyses (Figure 4C) confirmed the solid-like nature of all eutectozymes, with G′ consistently higher

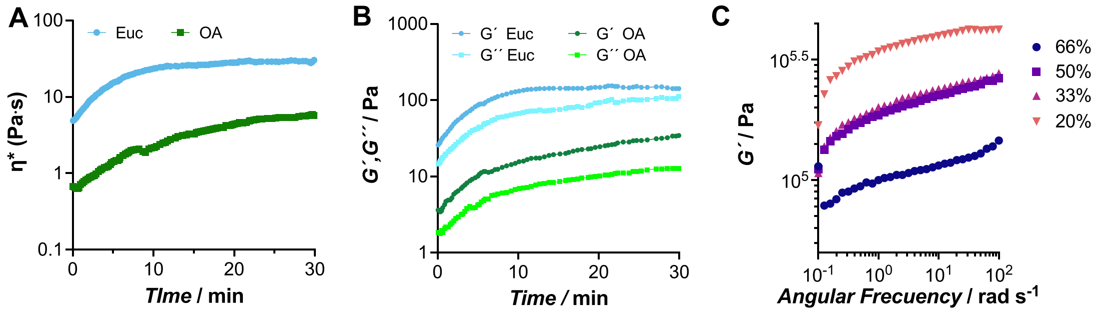


FIGURE 4 | Complex viscosity changes (A) and the evolution of the storage (G′) and loss (G′′) moduli over time (B) at constant frequency and amplitude in a viscosity-time assay for Euc and OA. The assays recorded viscosity and the dynamic moduli changes over 30 min, after which a plateau was observed. (C) Frequency sweeps of the eutectozymes at 20, 33, 50, and 66 wt.% of Men:Euc.

TABLE 2 | Biocatalytic parameters of GOx eutectozymes vs. free GOx.

Sample	$^{app}K_M$ [mM]	$^{app}k_{cat}$ [min^{-1}]	$^{app}k_{cat}/^{app}K_M$ [min^{-1} mM^{-1}]
OA@GOx	12.5 ± 0.8	463 ± 16	37.1
Euc@GOx	1.2 ± 0.6	872 ± 25	726.0
GOx free	25.0 ± 0.1	13446 ± 58	537.8

than the loss modulus (G″) across the 0.1–100 rad/s range [55, 56]. Low-HES gels exhibited minimal frequency dependence, consistent with covalent crosslinking dominating their viscoelastic response, whereas high-HES gels showed a slight increase in G′ at higher frequencies, indicative of slower relaxation processes in solvent-rich networks with weaker or reversible crosslinks [52, 55]. Overall, increasing the HES content from 20 to 66 wt.% led to a decrease in G′, consistent with the dilution of hydrophobic interactions and a lower effective crosslink density within the network [57]. These rheological features highlight the contribution of the dual-network architecture to the enhanced mechanical properties of eutectozymes [40–42, 50, 54, 58].

2.2.4 | Eutectozymes: Gel Materials That Support Biocatalytic Microreactors

Eutectozymes are presented herein as an innovative approach to enzyme immobilization. The presence of hydrophilic microcavities within the gel structure that presumably hold the loaded biocatalyst, i.e., GOx, may have a significant impact on the performance of GOx. Thus, the catalytic performance of the eutectozymes was monitored as described in the Supporting Information section. The resulting apparent kinetic parameters, calculated using the equations collected in Table S11, are summarized and compared to free GOx in Table 2.

As expected from heterogeneous biocatalysts, eutectozymes exhibited significant mass transfer limitations [59–61]. Eutectozymes show a drop of one order of magnitude in the apparent turnover frequency compared to free enzyme — $^{app}k_{cat}$ of ~13 × 10^3, 0.87 × 10^3, and 0.46 × 10^3 min^{-1} for free GOx, Euc@GOx, and OA@GOx, respectively (Figure S29). We ascribe such a drop to the confinement of the catalyst within the material, which imposes significant diffusion limitations on the system. This negative impact on turnover frequency has been observed in assorted biocatalytic hybrids where the enzyme is embedded within reticular materials. Hence, we previously reported a substantial decrease, compared to free enzyme, of ~25 times and 1.5–3.0 times in terms of $^{app}k_{cat}$ of GOx hybrids encapsulated in ZIF-8 frameworks and metal-organic enzyme aggregates (MOEAs), respectively [62]. Soft-crosslinked hydrogels also show crosslinker density-dependent substrate-transfer limitations that can result in a catalytic lag-phase and further protein conformational effects [63, 64].

Notably, the changes in $^{app}K_M$, a parameter directly correlated to enzyme-substrate affinity, were improved compared to free protein (Figure S30). This effect suggests that, although the HES and the gelification of the matrix may have imposed some conformational constraints on the protein, a higher affinity toward the substrate is observed, particularly for Euc@GOx ($^{app}K_M$ of 1.2 vs. 25.0 mM for Euc@GOx and free GOx, respectively). This effect is typical of confined environments, in which local substrate concentration, to which the enzyme is actually exposed, is significantly increased [47]. The hydrophobic nature of Euc@GOx might explain the formation of efficient microreactors in which both the enzyme and the substrate can accumulate, showing improved $^{app}K_M$ compared to gel-like materials such as alginate capsules — $^{app}K_M$ of 30 mM—[65], peptide hydrogels — $^{app}K_M$ of 3.1 mM—[66], or silica matrix — $^{app}K_M$ of 20.3 mM—[67].

2.2.5 | Eutectogels Confer Improved Features to the Enzyme

The use of heterogeneous supports is designed to enhance the stability of enzymes under harsh conditions, as well as to facilitate the reuse of proteins [68–70]. Therefore, we tested the resilience of the eutectozymes to harsh operational conditions. We subjected the eutectozymes to 60°C incubation for 30 min to test the resistance to thermal inactivation —free GOx collapses rapidly above ~50°C (Figure S31A,B). Under these conditions, Euc@GOx and OA@GOx retained ~82% and ~80% of their activity. Moreover, still ~28% and ~78% of the activity was retained by OA@GOx and Euc@GOx at 70°C, respectively (Figure 5A). The eutectozyme formulations conferred remarkably broad pH tolerance to GOx, representing a significant improvement over the native enzyme

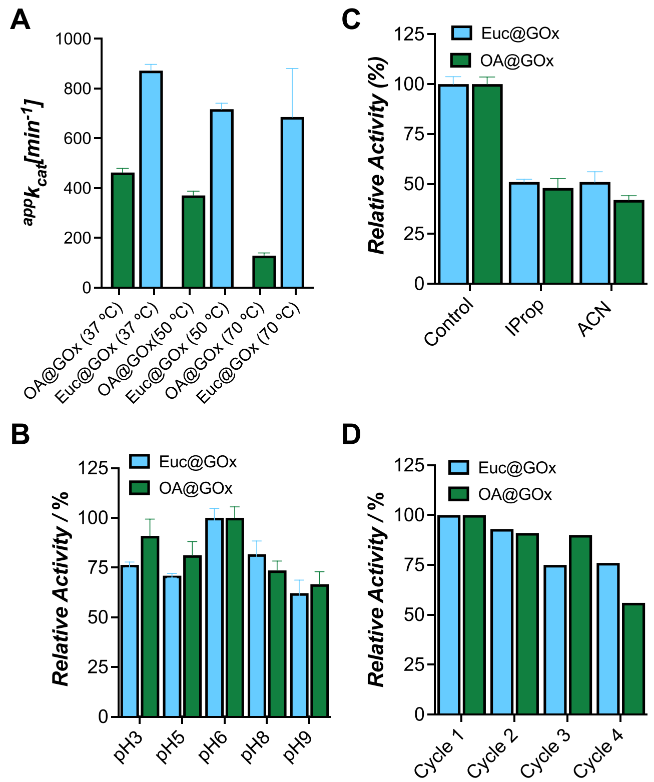


FIGURE 5 | Catalytic performance of Euc@GOx and OA@GOx under harsh operational conditions. (A) Remaining catalytic activity after incubating the eutectozymes at 37, 50, and 70°C for 30 min. (B) Measurement of the relative catalytic activity at broad pH conditions, from pH 3.0 to pH 9.0. (C) Remaining catalytic activity of the eutectozymes after being incubated for 1.5 h in 100% isopropanol (IProp) and acetonitrile (ACN). (D) Relative activity of Euc@GOx and OA@GOx after up to 4 reaction cycles. Results are expressed as mean ± standard deviation (n = 3).

(Figure S31C) [59]. While free GOx was essentially inactive at pH 3.0 or pH 9.0, the entrapped enzyme retained measurable activity even under these extreme conditions (Figure 5B) [71]. Further, the entrapped GOx demonstrated exceptional resistance to organic solvents that completely inactivate the free enzyme [59]. After 90 min in 100% (v/v) isopropanol or acetonitrile, both OA@GOx and Euc@GOx retained 40–60% activity, while free enzyme lost essentially all function (>80%) (Figure 5C; Figure S31D). With these results, we hypothesize that the microreactor cavities could retain a hydration shell and limit solvent access to the protein, preventing dehydration and unfolding [72]. Multicycle assays highlighted the practical durability and reusability of the eutectogel systems [59]. OA@GOx showed declining activity over repeated uses, but Euc@GOx maintained substantially more activity over multiple cycles. After four cycles, this HES retained approximately 76% of its initial activity, while OA@GOx declined to nearly 20% (Figure 5D).

These results demonstrate that the hydrophobic eutectogel creates a benign protective environment for GOx even in harsh solvent mixtures, consistent with the general strategy of immobilization to improve solvent tolerance [73]. Further, this multicomponent system creates an environment that combines the benefits of immobilized biocatalysts [74, 75] and HES, enhancing stability and recyclability.

2.3 | Advanced Applications of Eutectozymes

2.3.1 | Enzymatic Dye Degradation Using Euctectozymes: Efficiency and Environmental Applications

As a first approach to demonstrate the potential of eutectozymes, we introduce them as promising platforms for environmental remediation, particularly for the decontamination of aqueous effluents containing persistent aromatic compounds [76]. We previously demonstrated that the confinement of GOx with hemin, which was loaded as an HRP mimic, led to enhanced removal of polyphenols and recalcitrant dyes by just supplementing glucose to the media [77]. Here, we present a similar strategy in which both GOx and HRP are confined in the Men:Euc-based eutectogels (see Supporting Information for further details). Hence, eutectozymes were immersed in acid green 25 (AG25), methylene blue (MB), and Rho solutions (AG25: 16 mg L^{-1}; Rho: 40 mg L^{-1}; MB: 40 mg L^{-1}) in which glucose (200 mM) was supplemented in a sodium phosphate solution at pH 6.0 to conduct the catalysis. The decolorization of the solution was monitored at 610, 660, and 560 nm, respectively. As observed in Figure 6A,B, AG25 and MB, which have strong aromatic features and are known to be persistent in aqueous environments, showed significant absorbance decay by 2.5–3 h and continued fading up until the end of the analysis. Hence, after 10 h of incubation, eutectozymes achieved a decoloration of AG25 and MB of ≈ 86% and 73%, respectively (Figure S32). We observed that the degradation was pH-dependent, particularly in the case of AG25, showing up to 50% enhanced dye removal efficiencies at pH 7.5 (Figure S33). Finally, we also tested the degradation of Rho (Figure 6C), which was significantly faster, with ≈ 90% of the dye removed after 1 h of incubation. When comparing our system with other reported technologies for dye removal — including immobilized laccase systems, bacterial consortia, and single-enzyme matrices (Table S12)— the eutectozyme achieves competitive or superior degradation efficiencies (90–95%) within a shortened 10-hour timeframe under mild conditions, with notable recyclability. Unlike microbial systems and many traditional free or immobilized enzyme platforms, the eutectogel retains robust activity over a wide pH range (pH 5.0–7.5), providing a substantial operational advantage for water treatment applications subject to variable effluent chemistry. These results underscore the versatility and industrial promise of bienzyme eutectogels as advanced, sustainable materials for efficient dye remediation [76].

2.3.2 | Antimicrobial Properties of HES Eutectozymes

The antimicrobial properties of menthol and eucalyptol-based HES against *S. aureus* are well documented [78–81]. Moreover, we confirmed, using our experimental setup, that GOx and HRP show inhibitory or bactericidal effects against the methicillin-resistant *Staphylococcus aureus* (MRSA) ATCC 43300 (Figure S29A). GOx showed complete inhibition (measured as the minimum inhibitory concentration —MIC—) and killing (minimum bactericidal concentration —MBC—) of MRSA ATCC 43300 at higher concentrations than 0.0048 μM. This activity is attributed to its ability to generate H_2O_2 upon the oxidation of glucose [82]. The H_2O_2 damages bacterial cell membranes, proteins,

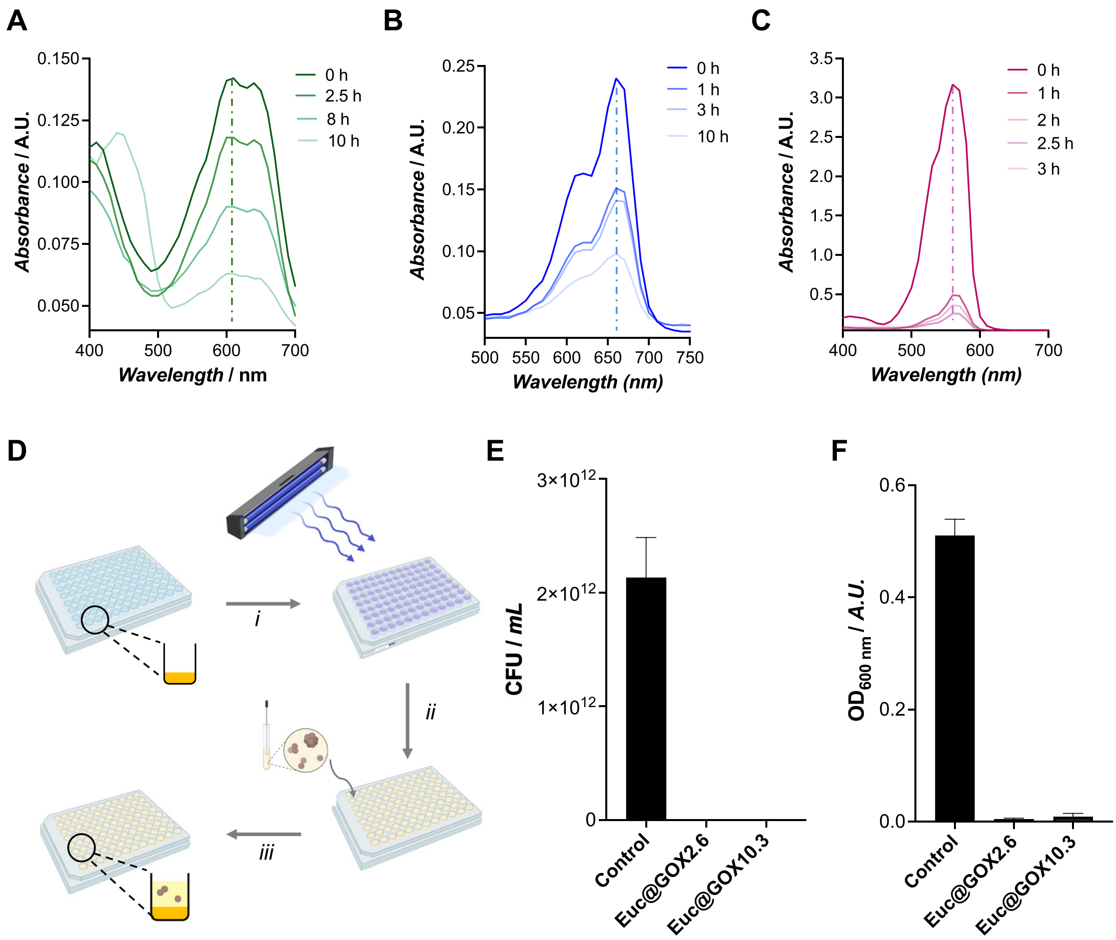


FIGURE 6 | Application of the eutectozymes for dye removal and as antimicrobial materials. The decolorization of AG-25 (A), MB (B), and Rho (C) was monitored at different time intervals upon the addition of glucose. (D) Schematic representation of the experimental setup used to evaluate the antimicrobial activity of eutectogels. The procedure setup entails the deposition of the eutectozymes and the UV sterilization of the material (i), the inoculation of the bacteria (ii), and the incubation and measurement of the bacterial growth in the wells (iii). (E) Inhibitory activity of eutectogels evaluated by measuring the optical density at 600 nm (OD_{600}) of bacterial suspensions incubated overnight at 37 °C with the eutectogels. (F) Bacterial viability of *S. aureus* after overnight incubation with the eutectogels at 37 °C measured by colony-forming unit (CFU) count. Results are expressed as mean ± standard deviation (n = 3).

and DNA, leading to cell lysis or growth inhibition [83–85]. We also assigned, to a lower extent, antimicrobial properties to the HRP, with both the MIC and MBC being 15.37 µM, several orders of magnitude higher than values measured for GOx. This activity is linked to its ability to catalyze oxidative reactions using various compounds present in the growth medium as substrates, generating reactive oxygen species (ROS) [86, 87]. Therefore, in the light of these results, we determined the antibacterial potential of the Euc@GOx, in which the synergistic antimicrobial action of the Men:Euc mixture and the enzymes may provide excellent antimicrobial properties.

Following the experimental setup illustrated in Figure 6D, eutectozymes were deposited on the bottom of the wells, and the growth of bacteria was monitored after 24 h of inoculation. To conduct these experiments, Euc@GOx with two different GOx loadings, i.e., 2.6 and 10.3 µg/µL, was tested. As shown in Figure 6E, both Euc@GOx formulations significantly inhibited bacterial growth. The bactericidal activity of the formulations was further evaluated by colony-forming unit (CFU) counting (Figure 6F) and by measuring bacterial metabolic activity using the Alamar Blue assay (Figure S33). Both experiments confirmed the complete removal of bacteria from the solution, revealing a great potential of the tested hybrids as antimicrobial materials. Such excellent antibacterial efficiency is comparable to other reported intricate GOx biohybrid systems in which radical oxygen species (ROS) are released —e.g., GOx/Ag nanoparticle hydrogels, GOx/MOF hydrogels, or GOx/MnO_2 polydopamine/acrylamide hydrogels—[88–90] Therefore, the combined action of the released hydrogen peroxide and the Euc eutectogel might be responsible for the strong antibacterial profile of Euc@GOx.

3 | Conclusion

This work positions eutectozymes as a new class of soft hybrid biocatalytic materials, bridging the advantages of supramolecular eutectogels and polymeric networks to address long-standing challenges in enzyme immobilization. Traditional immobilization strategies—ranging from rigid inorganic scaffolds such as MOFs, silica matrices, and nanoflowers to soft matrices like hydrogels or ionogels—often face trade-offs between enzyme activity retention, mechanical robustness, and recyclability. In contrast, eutectozymes overcome these limitations by merging natural hydrophobic eutectic solvents (HES) with a dual supramolecular–polymeric network, providing a confined yet adaptive microenvironment that preserves enzyme conformation and stability while maintaining structural integrity under stress.

Regarding the design of the eutectogels, natural HES based on menthol mixtures were selected as good matrix candidates due to their compatibility with GOx, and the excellent material integrity and mechanical properties shown in aqueous media. Yet, the assembly into supramolecular gels is challenging since the supramolecular network, while offering excellent dynamic properties, lacks sufficient mechanical strength or structural integrity for long-term stability under stress. Conversely, the covalent polymer network was unable to impart sufficient structural integrity. Likely, harsher conditions, such as increased temperature or sonication pulses, could enhance reticular network formation; however, since the in situ polymerization process is performed in the presence of the enzyme, GOx in this case, the use of mild conditions is required. Therefore, in light of our results, the gelification of these menthol-based HES requires the formation of a double network that yields gel-like materials. In this research, we have demonstrated the importance of oxidation in the DBS gelator for generating gel-like materials using the herein tested systems.

The synthetic procedure yielded highly structured eutectogels, both at the nano- and microscale. While SAXS and WAXS revealed the formation of lamellar organization of supramolecular and crystalline phases, SEM images suggested the formation of microfibers. Such a structure is conserved in the eutectozymes, although a higher amorphous content was detected at high protein loadings. Interestingly, the materials contain microcavities, likely formed through liquid-liquid phase separation between the hydrophobic matrix and the residual water introduced into the reaction medium. The ultrafast gelification, less than 1 min for the Euc@GOx formulation, freezes the formed emulsion, preserving the hydrophilic microcavities that are converted into microreactors in the presence of the enzyme. Our findings demonstrate that eutectozymes exhibit remarkable stability at broad operational conditions, i.e., high temperatures, extreme pH, and the presence of organic solvents, overperforming the activity of free GOx. The high reusability of these materials, particularly those fabricated with Men:Euc mixture, makes them excellent candidates as effective biocatalytic platforms. Moreover, beyond catalytic recycling, the simplicity and low cost of the HES components, together with their facile preparation by direct mixing, support the scalability of these systems. Although long-term storage stability remains to be systematically evaluated, the low volatility and chemically protective hydrogen-bonded microenvironment characteristic of eutectic media are expected to mitigate enzyme denaturation and microbial degradation over time, suggesting favorable prospects for industrial deployment.

Functionally, eutectozymes demonstrated improved substrate affinity (lower appKM), operational resilience, and reusability, highlighting their potential as sustainable, high-performance biocatalysts. Their versatility was further exemplified in environmental remediation and antimicrobial applications, where they achieved rapid dye degradation (>90%) and potent bactericidal effects, surpassing most existing enzymatic or antimicrobial platforms.

Placed within the landscape of sustainable hybrid biocatalytic materials, eutectozymes uniquely integrate green chemistry principles with advanced material design: they employ biocompatible, low-toxicity components, mild synthesis routes, and recyclable architectures, offering a scalable pathway toward soft, adaptive, and multifunctional biocatalytic systems. Further, while our initial demonstration focused on HES due to their enhanced stability in aqueous environments, the eutectozyme concept is readily extensible to other eutectic systems, thereby positioning them as versatile, highly effective biocatalytic platforms with great potential beyond just enzyme immobilization. As a result, we anticipate the implementation of innovative sustainable enzyme-based technologies across fields such as bioelectronics, biosensing, and soft-actuating devices.

Author Contributions

Manuel Eduardo Martinez Cartagena conceptualized the idea for the study; designed the methodology; performed investigation and data curation; wrote the original draft; and reviewed and edited the final manuscript. Lucia Suarez designed the methodology and performed investigation on material preparation. Aitor Ontoria designed the methodology and performed the investigation on material characterization. Francisca J. Benitez conceptualized the idea for the study and performed the formal analysis, investigation, and visualization of the computational chemistry calculations. Elixabete Rezabal conceptualized the idea for the study and performed the formal analysis, investigation, and visualization of the computational chemistry calculations. María Soledad Orellano and Marcelo Calderón conceptualized the idea for the study; performed the investigation; and wrote and reviewed the final manuscript. Cristián Huck-Iriart and Agustin S. Picco conceptualized the idea for the study; performed the investigation and characterization WAXS and SAXS; and wrote and reviewed the final manuscript. Matias L. Picchio and Ana Beloqui conceptualized the idea for the study; performed the investigation, supervision, and funding acquisition; wrote, reviewed, and edited the final manuscript.

Acknowledgements

A.B., M.L.P., and M.C. are thankful to the IKERBASQUE-Basque Foundation for Science. A.B. and M.L.P. thank the Ramón Areces Foundation for the financial support received through the “XXII Concurso Nacional para la adjudicación de Ayudas a la Investigación en Ciencias de la Vida y de la Materia” (CIVP22S18625). A.B. thanks the University of the Basque Country (GIU21/033), the Spanish Research Agency (PID2022-142128NB-I00 and CNS2023-145416 funded by MCIN/ AEI/10.13039/501100011033/ and the “European Union NextGenerationEU/PRTR”; “María de Maeztu” Programme for Center of Excellence in R&D, grant CEX2023-001303-M funded by MICIU/AEI/ 10.13039/501100011033). M.L.P. gratefully acknowledges the financial support from the Basque Government, Department of Education (IT1766-22). M.C. has received funding from the University of the Basque Country (projects COLLAB22/05 and GIU21/033). This research was funded by the European Union under the IONBIKE 2.0 project (grant agreement n. 101129945, https://doi.org/10.3030/101129945). A.S.P thanks UNLP and CONICET for their support. A.S.P is staff member of CONICET. C.H.-I. acknowledges ALBA Synchrotron Light source, Barcelona, Spain (project ID 2023027439). E.R. and F.J.B. acknowledge HPCAI16: MLFF4Bio (IKUR), IT-1584-22 project of the Basque Government, and the Technical and human support provided by IZO-SGI, SGIker (UPV/EHU, MICINN, GV/EJ, ERDF and ESF) and the DIPC Supercomputing

Center. AO thanks grant ayuda Predoctoral de Formación de Personal Investigador no Doctor, Departamento de Ciencia, Universidades e Innovación Eusko Jaurlaritza-Gobierno Vasco (PRE_2023_1_0235). The authors thank CEDRO Beamline of the Brazilian Synchrotron Light Laboratory (LNLS-Sirius, Brasil) for the preliminary circular dichroism experiments (proposal ID: 20241888). Open Access funding enabled by EHU.

Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. Y. R. Maghraby, R. M. El-Shabasy, A. H. Ibrahim, and H. M. E. S. Azzazy, "Enzyme Immobilization Technologies and Industrial Applications," *ACS Omega* 8 (2023): 5184–5196, https://doi.org/10.1021/ACSOMEGA.2C07560.

2. J. M. Bolivar, J. M. Woodley, and R. Fernandez-Lafuente, "Is Enzyme Immobilization a Mature Discipline? Some Critical Considerations to Capitalize on the Benefits of Immobilization," *Chemical Society Reviews* 51 (2022): 6251–6290, https://doi.org/10.1039/D2CS00083K.

3. K. Y. Wang, J. Zhang, Y. C. Hsu, et al., "Bioinspired Framework Catalysts: From Enzyme Immobilization to Biomimetic Catalysis," *Chemical Reviews* 123 (2023): 5347–5420, https://doi.org/10.1021/ACS.CHEMREV.2C00879.

4. A. Huang, Z. W. Li, L. Guo, et al., "An Aggregation-Induced Conformation Locking Strategy Facilitates the Activation of Lipase Biocatalyst," *Nature Communications* 16, no. 1 (2025): 1–13, https://doi.org/10.1038/s41467-025-59824-w.

5. J. Zhang, J. C. White, G. V. Lowry, et al., "Advanced Enzyme-Assembled Hydrogels for the Remediation of Contaminated Water," *Nature Communications* 16, no. 1 (2025): 1–11, https://doi.org/10.1038/s41467-025-58338-9.

6. G. Zheng, Z. Zhou, J. Yang, et al., "Robust Immobilization and Activity Preservation of Enzymes in Porous Frameworks by Silica-Based "Inorganic Glue"" *Advanced Materials* 36 (2025): 2407831, https://doi.org/10.1002/ADMA.202407831.

7. L. Ran, Y. Lu, L. Chen, M. He, and Z. Deng, "Design, Synthesis, and Application of Immobilized Enzymes on Artificial Porous Materials," *Advanced Science* 12 (2025): 2500345, https://doi.org/10.1002/ADVS.202500345.

8. Y. Feng, R. Shi, M. Yang, Y. Zheng, Z. Zhang, and Y. Chen, "A Dynamic Defect Generation Strategy for Efficient Enzyme Immobilization in Robust Metal-Organic Frameworks for Catalytic Hydrolysis and Chiral Resolution," *Angewandte Chemie International Edition* 62 (2023): 202302436, https://doi.org/10.1002/ANIE.202302436.

9. P. D. Patil, R. K. Kelkar, N. P. Patil, et al., "Magnetic Nanoflowers: A Hybrid Platform for Enzyme Immobilization," *Critical Reviews in Biotechnology* 44 (2024): 795–816, https://doi.org/10.1080/07388551.2023.2230518.

10. Y. Yuan, J. Shen, and S. Salmon, "Developing Enzyme Immobilization With Fibrous Membranes: Longevity and Characterization Considerations," *Membranes (Basel)* 13 (2023): 532, https://doi.org/10.3390/MEMBRANES13050532/S1.

11. N. Zeballos, N. Comino, D. Andrés-Sanz, et al., "Region-Directed Enzyme Immobilization Through Engineering Protein Surface With Histidine Clusters," *ACS Appl Mater Interfaces* 16 (2024): 833–846, https://doi.org/10.1021/ACSAMI.3C15993/SUPPL_FILE/AM3C15993_SI_001.PDF.

12. J. Meyer, L. E. Meyer, and S. Kara, "Enzyme Immobilization in Hydrogels: A Perfect Liaison for Efficient and Sustainable Biocatalysis," *Engineering in Life Sciences* 22 (2022): 165–177, https://doi.org/10.1002/ELSC.202100087.

13. O. N. Makshakova, L. R. Bogdanova, A. O. Makarova, et al., "κ-Carrageenan Hydrogel as a Matrix for Therapeutic Enzyme Immobilization," *Polymers* 14 (2022): 4071, https://doi.org/10.3390/POLYM14194071.

14. V. B. Juska and M. E. Pemble, "A Critical Review of Electrochemical Glucose Sensing: Evolution of Biosensor Platforms Based on Advanced Nanosystems," *Sensors* 20 (2020): 6013, https://doi.org/10.3390/S20216013.

15. C. Manasa, V. Basavanna, and S. Ningaiah, "Ionic Liquid-Based Hybrid Materials: Ionogel Review," *Biointerface Research in Applied Chemistry* 13 (2023): 391, https://doi.org/10.33263/BRIAC134.391.

16. M. Criado-Gonzalez, N. Alegret, A. M. Fracaroli, et al., "Mixed Conductive, Injectable, and Fluorescent Supramolecular Eutectogel Composites," *Angewandte Chemie International Edition* 62 (2023): 202301489, https://doi.org/10.1002/ANIE.202301489.

17. C. Esteves and A. C. A. Roque, "Short Peptide and Amino Acid-Based Supramolecular Ionogels and Eutectogels," *Chemistry (Weinheim An Der Bergstrasse, Germany)* 30 (2024): 202400622, https://doi.org/10.1002/CHEM.202400622.

18. J. Ruiz-Olles, P. Slavik, N. K. Whitelaw, and D. K. Smith, "Self-Assembled Gels Formed in Deep Eutectic Solvents: Supramolecular Eutectogels With High Ionic Conductivity," *Angewandte Chemie International Edition* 58 (2019): 4173–4178, https://doi.org/10.1002/ANIE.201810600.

19. M. L. Picchio, D. Minudri, D. Mantione, et al., "Natural Deep Eutectic Solvents Based on Choline Chloride and Phenolic Compounds as Efficient Bioadhesives and Corrosion Protectors," *ACS Sustainable Chemistry & Engineering* 10 (2022): 8135–8142, https://doi.org/10.1021/ACSSUSCHEMENG.2C01976/ASSET/IMAGES/LARGE/SC2C01976_0006.JPEG.

20. K. K. Arjunan, C. Y. Weng, Y. J. Sheng, and H. K. Tsao, "Formation of Self-Healing Granular Eutectogels Through Jammed Carbopol Microgels in Supercooled Deep Eutectic Solvent," *Langmuir* 40 (2024): 17081–17089, https://doi.org/10.1021/ACS.LANGMUIR.4C02069.

21. C. Zeng, H. Zhao, Z. Wan, Q. Xiao, H. Xia, and S. Guo, "Highly Biodegradable, Thermostable Eutectogels Prepared by Gelation of Natural Deep Eutectic Solvents Using Xanthan Gum: Preparation and Characterization," *RSC Advances* 10 (2020): 28376–28382, https://doi.org/10.1039/D0RA03390A.

22. K. Kumar, M. Calderón, A. Beloqui, and M. L. Picchio, "Eutectogels as Promising Materials in Biocatalysis," *Chemcatchem* 16 (2024): 202400204, https://doi.org/10.1002/CCTC.202400204.

23. P. Makoś, E. Słupek, and A. Małachowska, "Silica Gel Impregnated by Deep Eutectic Solvents for Adsorptive Removal of BTEX From Gas Streams," *Materials* 13 (2020): 1894, https://doi.org/10.3390/MA13081894.

24. P. A. Mercadal, A. González, A. Beloqui, et al., "Eutectogels: The Multifaceted Soft Ionic Materials of Tomorrow," *JACS Au* 4 (2024): 3744–3758, https://doi.org/10.1021/JACSAU.4C00677/ASSET/IMAGES/LARGE/AU4C00677_0006.JPEG.

25. S. Marullo, F. Petta, G. Infurna, N. T. Dintcheva, and F. D'Anna, "Polysaccharide-Based Supramolecular Bicomponent Eutectogels as Sustainable Antioxidant Materials," *Green Chemistry* 25 (2023): 4513–4527, https://doi.org/10.1039/D3GC00573A.

26. B. Saavedra, A. Meli, C. Rizzo, D. J. Ramón, and F. D'Anna, "Natural Eutectogels: Sustainable Catalytic Systems for C–C Bond Formation Reactions," *Green Chemistry* 23 (2021): 6555–6565, https://doi.org/10.1039/D1GC01647D.

27. E. P. Carreiro, H. J. Federsel, G. J. Hermann, and A. J. Burke, "Stereoselective Catalytic Synthesis of Bioactive Compounds in Natural Deep Eutectic Solvents (NADESs): A Survey Across the Catalytic Spectrum," *Catalysts* 14 (2024): 160, https://doi.org/10.3390/CATAL14030160.

28. O. Długosz, “Natural Deep Eutectic Solvents in the Synthesis of Inorganic Nanoparticles,” *Materials* 16 (2023): 627, https://doi.org/10.3390/MA16020627.

29. C. Ferreira, M. A. Sarraguça, C. Ferreira, and M. Sarraguça, “A Comprehensive Review on Deep Eutectic Solvents and Its Use to Extract Bioactive Compounds of Pharmaceutical Interest,” *Pharmaceuticals* 17 (2024): 124, https://doi.org/10.3390/PH17010124.

30. J. Guo, N. Zhao, Y. Zhao, et al., “Effect of Deep Eutectic Solvents on the Activity and Stability of Cellulases and Pectinases,” *ACS Omega* 8 (2023): 45678–45686, https://doi.org/10.1021/ACSOMEGA.3C06088.

31. A. Sanchez-Fernandez, J. F. Poon, A. E. Leung, S. F. Prévost, and C. Dicko, “Stabilization of Non-Native Folds and Programmable Protein Gelation in Compositionally Designed Deep Eutectic Solvents,” *ACS Nano* 18 (2024): 18314–18326, https://doi.org/10.1021/ACSNANO.4C01950/SUPPL_FILE/NN4C01950_SI_001.PDF.

32. N. Zhang, V. Lahmann, J. P. Bittner, et al., “Redox Biocatalysis in Lidocaine-Based Hydrophobic Deep Eutectic Solvents: Non-Conventional Media Outperform Aqueous Conditions,” *Chemsuschem* 18 (2025): 202402075, https://doi.org/10.1002/CSSC.202402075.

33. V. Nayak, D. D, G. Jude Martis, and S. L. Gaonkar, “Advancements in Biocatalysis: A Comprehensive Review of Applications in Organic Synthesis,” *Synthetic Communications* 55 (2025): 443–464, https://doi.org/10.1080/00397911.2024.2449258.

34. P. Xu, G. W. Zheng, M. H. Zong, N. Li, and W. Y. Lou, “Recent Progress on Deep Eutectic Solvents in Biocatalysis,” *Bioresources and Bioprocessing* 4 (2017): 1–18, https://doi.org/10.1186/S40643-017-0165-5/METRICS.

35. N. Zhang, P. Domínguez de María, and S. Kara, “Biocatalysis for the Synthesis of Active Pharmaceutical Ingredients in Deep Eutectic Solvents: State-Of-The-Art and Prospects,” *Catalysts* 14 (2024): 84, https://doi.org/10.3390/CATAL14010084.

36. Y. Alacid, R. Martínez-Baquero, R. Esquembre, F. Montilla, M. J. Martínez-Tomé, and C. R. Mateo, “Innovative Fluorescent Nanocomposite Eutectogels: Design and Characterization Towards Biosensing Applications,” *Journal of Molecular Liquids* 424 (2025): 127123, https://doi.org/10.1016/J.MOLLIQ.2025.127123.

37. E. L. Smith, A. P. Abbott, and K. S. Ryder, “Deep Eutectic Solvents (DESs) and Their Applications,” *Chemical Reviews* 114 (2014): 11060–11082, https://doi.org/10.1021/CR300162P/ASSET/IMAGES/LARGE/CR-2012-00162P_0004.JPEG.

38. J. L. de Lacalle, M. L. Picchio, A. Dominguez-Alfaro, et al., “Hydrophobic Eutectogels as Electrodes for Underwater Electromyography Recording,” *ACS Materials Letters* 5 (2023): 3340–3346, https://doi.org/10.1021/ACSMATERIALSLETT.3C00938/ASSET/IMAGES/LARGE/TZ3C00938_0004.JPEG.

39. U. A. Weerasinghe, T. Wu, P. L. Chee, et al., “Deep Eutectic Solvents Towards Green Polymeric Materials,” *Green Chemistry* 26 (2024): 8497–8527, https://doi.org/10.1039/D4GC00532E.

40. C. Loebel, A. Ayoub, J. H. Galarraga, et al., “Tailoring Supramolecular Guest–Host Hydrogel Viscoelasticity With Covalent Fibrinogen Double Networks,” *Journal of Materials Chemistry B* 7 (2019): 1753–1760, https://doi.org/10.1039/C8TB02593B.

41. P. R. A. Chivers and D. K. Smith, “Spatially-resolved Soft Materials for Controlled Release—Hybrid Hydrogels Combining a Robust Photo-Activated Polymer Gel With an Interactive Supramolecular Gel,” *Chemical Science* 8 (2017): 7218–7227, https://doi.org/10.1039/C7SC02210G.

42. K. Nakamura, R. Kubota, T. Aoyama, K. Urayama, and I. Hamachi, “Four Distinct Network Patterns of Supramolecular/Polymer Composite Hydrogels Controlled by Formation Kinetics and Interfiber Interactions,” *Nature Communications* 14, no. 1 (2023): 1–12, https://doi.org/10.1038/s41467-023-37412-0.

43. B. O. Okesola, V. M. P. Vieira, D. J. Cornwell, N. K. Whitelaw, and D. K. Smith, “1,3:2,4-Dibenzylidene- d -Sorbitol (DBS) and Its Derivatives—Efficient, Versatile and Industrially-Relevant Low-Molecular-Weight Gelators With Over 100 Years of History and a Bright Future,” *Soft Matter* 11 (2015): 4768–4787, https://doi.org/10.1039/C5SM00845J.

44. X. Jiang, X. Chen, C. Ling, S. Chen, and Z. Wu, “High-Performance Cu/ZnO Catalysts Prepared Using a Three-Channel Microreactor,” *Applied Catalysis A: General* 570 (2019): 192–199, https://doi.org/10.1016/J.APCATA.2018.11.023.

45. S. Parvate, P. Dixit, and S. Chattopadhyay, “Superhydrophobic Surfaces: Insights From Theory and Experiment,” *Journal of Physical Chemistry B* 124 (2020): 1323–1360, https://doi.org/10.1021/ACS.JPCB.9B08567/ASSET/IMAGES/MEDIUM/JP9B08567_M040.GIF.

46. N. K. McLeod, L. Stokes, J. Lewis, and D. K. Smith, “Supramolecular Gels With Potential Applications as Anti-Icing Agents,” *Langmuir* 41 (2025): 13932–13947, https://doi.org/10.1021/ACS.LANGMUIR.5C00755.

47. A. Singh, F. I. Auzanneau, M. G. Corradini, G. Grover, R. G. Weiss, and M. A. Rogers, “Molecular Nuances Governing the Self-Assembly of 1,3:2,4-Dibenzylidene- d -Sorbitol,” *Langmuir* 33 (2017): 10907–10916, https://doi.org/10.1021/ACS.LANGMUIR.7B02191/SUPPL_FILE/LA7B02191_SI_001.PDF.

48. M. Ambrosi, M. Raudino, I. Diañez, and I. Martínez, “Non-Isothermal Crystallization Kinetics and Morphology of Poly(3-Hydroxybutyrate)/Pluronic Blends,” *European Polymer Journal* 120 (2019): 109189, https://doi.org/10.1016/J.EURPOLYMJ.2019.08.016.

49. S. Nojima, Y. Terashima, and T. Ashida, “Small-Angle X-Ray Scattering Study of the Morphology of Blends of Poly(ε-Caprolactone) and Polystyrene Oligomer,” *Polymer (Guildf)* 27 (1986): 1007–1013, https://doi.org/10.1016/0032-3861(86)90064-9.

50. H. Sun, B. Zhang, L. Lu, et al., “d-Gluconic Acetal Gelator-Based Supramolecular—Polymer Dual Network Eutectogels for High Performance Temperature, Strain, and Pressure Sensors,” *Chemical Engineering Journal* 451 (2023): 139051, https://doi.org/10.1016/J.CEJ.2022.139051.

51. D. Wang, X. Yang, Q. Liu, L. Yu, and J. Ding, “Enzymatically Cross-Linked Hydrogels Based on a Linear Poly(ethylene glycol) Analogue for Controlled Protein Release and 3D Cell Culture,” *Journal of Materials Chemistry B* 6 (2018): 6067–6079, https://doi.org/10.1039/C8TB01949E.

52. C. López-Serrano, Y. Côté-Paradis, B. Habenstein, et al., “Integrating Mechanics and Bioactivity: A Detailed Assessment of Elasticity and Viscoelasticity at Different Scales in 2D Biofunctionalized PEGDA Hydrogels for Targeted Bone Regeneration,” *ACS Appl Mater Interfaces* 16 (2024): 39165–39180, https://doi.org/10.1021/ACSAMI.4C10755/SUPPL_FILE/AM4C10755_SI_001.PDF.

53. Y. Liang, K. Wang, J. Li, et al., “Low-Molecular-Weight Supramolecular-Polymer Double-Network Eutectogels for Self-Adhesive and Bidirectional Sensors,” *Advanced Functional Materials* 31 (2021): 2104963, https://doi.org/10.1002/ADFM.202104963.

54. K. Wang, H. Wang, J. Li, et al., “Super-Stretchable and Extreme Temperature-Tolerant Supramolecular-Polymer Double-Network Eutectogels With Ultrafast in Situ Adhesion and Flexible Electrochromic Behaviour,” *Materials Horizons* 8 (2021): 2520–2532, https://doi.org/10.1039/D1MH00725D.

55. G. Stojkov, Z. Niyazov, F. Picchioni, and R. K. Bose, “Relationship Between Structure and Rheology of Hydrogels for Various Applications,” *Gels* 7 (2021): 255, https://doi.org/10.3390/GELS7040255.

56. K. Alam, M. Iqbal, A. Hasan, and N. Al-Maskari, “Rheological Characterization of Biological Hydrogels in Aqueous State,” *Journal of Applied Biotechnology Reports* 7 (2020): 171–175, https://doi.org/10.30491/JABR.2020.109994.

57. C. Zeng, H. Zhao, Z. Wan, Q. Xiao, H. Xia, and S. Guo, “Highly Biodegradable, Thermostable Eutectogels Prepared by Gelation of Natural Deep Eutectic Solvents Using Xanthan Gum: Preparation and Characterization,” *RSC Advances* 10 (2020): 28376–28382, https://doi.org/10.1039/D0RA03390A.

58. C. B. Rodell, N. N. Dusaj, C. B. Highley, and J. A. Burdick, “Injectable and Cytocompatible Tough Double-Network Hydrogels Through Tandem

Supramolecular and Covalent Crosslinking," *Advanced Materials* 28 (2016): 8419–8424, https://doi.org/10.1002/ADMA.201602268.

59. M. Hümmer, S. Kara, A. Liese, I. Huth, J. Schrader, and D. Holtmann, "Synthesis of (-)-Menthol Fatty Acid Esters in and From (-)-Menthol and Fatty Acids—Novel Concept for Lipase Catalyzed Esterification Based on Eutectic Solvents," *Molecular Catalysis* 458 (2018): 67–72, https://doi.org/10.1016/j.mcat.2018.08.003.

60. M. D. Gouda, M. S. Thakur, and N. G. Karanth, "Reversible Denaturation Behavior of Immobilized Glucose Oxidase," *Applied Biochemistry and Biotechnology* 102-103 (2002): 471–480, https://doi.org/10.1385/ABAB:102-103:1-6:471.

61. Y. R. Maghraby, R. M. El-Shabasy, A. H. Ibrahim, and H. M. E. S. Azzazy, "Enzyme Immobilization Technologies and Industrial Applications," *ACS Omega* 8 (2023): 5184–5196, https://doi.org/10.1021/ACSOMEGA.2C07560/ASSET/IMAGES/LARGE/AO2C07560_0003.JPEG.

62. Y. Weng, R. Chen, Y. Hui, D. Chen, and C. X. Zhao, "Boosting Enzyme Activity in Enzyme Metal–Organic Framework Composites," *Chem and Bio Engineering* 1 (2024): 99–112, https://doi.org/10.1021/CBE.3C00091/ASSET/IMAGES/LARGE/BE3C00091_0005.JPEG.

63. E. Jang, S. Park, S. Park, et al., "Fabrication of Poly(Ethylene Glycol)-Based Hydrogels Entrapping Enzyme-Immobilized Silica Nanoparticles," *Polymers for Advanced Technologies* 21 (2010): 476–482, https://doi.org/10.1002/PAT.1455.

64. H. Laurent, D. J. Brockwell, and L. Dougan, "Nanomachine Networks: Functional All-Enzyme Hydrogels From Photochemical Cross-Linking of Glucose Oxidase," *Biomacromolecules* 26 (2025): 1195–1206, https://doi.org/10.1021/ACS.BIOMAC.4C01519/ASSET/IMAGES/LARGE/BM4C01519_0008.JPEG.

65. A. Blandino, M. Macías, and D. Cantero, "Immobilization of Glucose Oxidase Within Calcium Alginate Gel Capsules," *Process Biochemistry* 36 (2001): 601–606, https://doi.org/10.1016/S0032-9592(00)00240-5.

66. J. H. Kim, S. Y. Lim, D. H. Nam, J. Ryu, S. H. Ku, and C. B. Park, "Self-Assembled, Photoluminescent Peptide Hydrogel as a Versatile Platform for Enzyme-Based Optical Biosensors," *Biosensors & Bioelectronics* 26 (2011): 1860–1865, https://doi.org/10.1016/J.BIOS.2010.01.026.

67. W. Z. Jia, K. Wang, Z. J. Zhu, H. T. Song, and X. H. Xia, "One-Step Immobilization of Glucose Oxidase in a Silica Matrix on a Pt Electrode by an Electrochemically Induced Sol–Gel Process," *Langmuir* 23 (2007): 11896–11900, https://doi.org/10.1021/LA7020269/ASSET/IMAGES/LARGE/LA7020269F00008.JPEG.

68. A. Rodriguez-Abetxuko, D. Sánchez-deAlcázar, P. Muñumer, and A. Beloqui, "Tunable Polymeric Scaffolds for Enzyme Immobilization," *Frontiers in Bioengineering and Biotechnology* 8 (2020): 554323, https://doi.org/10.3389/FBIOE.2020.00830/XML.

69. J. Morgenstern, G. Gil Alvaradejo, N. Bluthardt, A. Beloqui, G. Delaittre, and J. Hubbuch, "Impact of Polymer Bioconjugation on Protein Stability and Activity Investigated With Discrete Conjugates: Alternatives to PEGylation," *Biomacromolecules* 19 (2018): 4250–4262, https://doi.org/10.1021/ACS.BIOMAC.8B01020/SUPPL_FILE/BM8B01020_SI_001.PDF.

70. M. Heredero and A. Beloqui, "Enzyme-Polymer Conjugates for Tuning, Enhancing, and Expanding Biocatalytic Activity," *Chembiochem* 24 (2023): 202200611, https://doi.org/10.1002/CBIC.202200611.

71. J. A. Bauer, M. Zámocká, J. Majtán, and V. Bauerová-Hlinková, "Glucose Oxidase, an Enzyme "Ferrari": Its Structure, Function, Production and Properties in the Light of Various Industrial and Biotechnological Applications," *Biomolecules* 12 (2022): 472, https://doi.org/10.3390/BIOM12030472.

72. Z. Nasir, A. Ali, M. F. Alam, M. Shoeb, and S. Nusrat Jahan, "Immobilization of GOx Enzyme on SiO_2 -Coated Ni–Co Ferrite Nanocomposites as Magnetic Support and Their Antimicrobial and Photocatalytic Activities," *ACS Omega* 6 (2021): 33554–33567, https://doi.org/10.1021/ACSOMEGA.1C04360/ASSET/IMAGES/MEDIUM/AO1C04360_M003.GIF.

73. M. S. Álvarez, M. A. Longo, A. Rodríguez, and F. J. Deive, "The Role of Deep Eutectic Solvents in Catalysis. A Vision on Their Contribution to Homogeneous, Heterogeneous and Electrocatalytic Processes," *Journal of Industrial and Engineering Chemistry* 132 (2024): 36–49, https://doi.org/10.1016/J.JIEC.2023.11.030.

74. A. Beloqui, S. Baur, V. Trouillet, et al., "Single-Molecule Encapsulation: A Straightforward Route to Highly Stable and Printable Enzymes," *Small* 12 (2016): 1716–1722, https://doi.org/10.1002/SMLL.201503405.

75. C. Coscolín, A. Beloqui, M. Martínez-Martínez, et al., "Controlled Manipulation of Enzyme Specificity Through Immobilization-Induced Flexibility Constraints," *Appl Catal A Gen* 565 (2018): 59–67, https://doi.org/10.1016/J.APCATA.2018.08.003.

76. N. Šekuljica, N. Prlainović, A. B. Stefanović, et al., "Decolorization of Anthraquinonic Dyes From Textile Effluent Using Horseradish Peroxidase: Optimization and Kinetic Study," *Scientific World Journal* 2015 (2015): 371625, https://doi.org/10.1155/2015/371625.

77. A. Rodriguez-Abetxuko, P. Muñumer, M. Okuda, J. Calvo, M. Knez, and A. Beloqui, "Nanoconfined (Bio)Catalysts as Efficient Glucose-Responsive Nanoreactors," *Advanced Functional Materials* 30 (2020): 2002990, https://doi.org/10.1002/ADFM.202002990.

78. F. Oliveira, E. Silva, A. Matias, J. M. Silva, R. L. Reis, and A. R. C. Duarte, "Menthol-Based Deep Eutectic Systems as Antimicrobial and Anti-Inflammatory Agents for Wound Healing," *European Journal of Pharmaceutical Sciences* 182 (2023): 106368, https://doi.org/10.1016/J.EJPS.2022.106368.

79. D. Kifer, V. Mužinić, and M. Š. Klaric, "Antimicrobial Potency of Single and Combined Mupirocin and Monoterpenes, Thymol, Menthol and 1,8-cineole Against Staphylococcus Aureus Planktonic and Biofilm Growth," *The Journal of Antibiotics* 69, no. 9 (2016): 689–696, https://doi.org/10.1038/ja.2016.10.

80. E. Silva, F. Oliveira, J. M. Silva, R. L. Reis, and A. R. C. Duarte, "Untangling the Bioactive Properties of Therapeutic Deep Eutectic Solvents Based on Natural Terpenes," *Current Research in Chemical Biology* 1 (2021): 100003, https://doi.org/10.1016/J.CRCHBI.2021.100003.

81. A. Viñas-Ospino, I. Sá-Nogueira, A. R. Duarte, et al., "Exploring the Biological Properties and Bioaccessibility of Orange Peel Extracts Using Deep Eutectic Systems," *Food Bioscience* 61 (2024): 104684, https://doi.org/10.1016/J.FBIO.2024.104684.

82. T. Zhang, Y. Qu, P. A. Gunatillake, P. Cass, K. E. S. Locock, and L. D. Blackman, "Honey-Inspired Antimicrobial Hydrogels Resist Bacterial Colonization Through Twin Synergistic Mechanisms," *Scientific Reports* 10, no. 1 (2020): 1–9, https://doi.org/10.1038/s41598-020-72478-6.

83. M. Tiina and M. Sandholm, "Antibacterial Effect of the Glucose Oxidase-Glucose System on Food-Poisoning Organisms," *International Journal of Food Microbiology* 8 (1989): 165–174, https://doi.org/10.1016/0168-1605(89)90071-8.

84. B. Zhao, M. Yuan, L. Wang, et al., "Antibacterial Activity of Bifunctional Bacterial Cellulose Composite Grafted With Glucose Oxidase and L-Arginine," *Cellulose* 30 (2023): 8973–8984, https://doi.org/10.1007/S10570-023-05406-2/FIGURES/6.

85. I. Lee, H. J. Cheon, M. D. Adhikari, et al., "Glucose Oxidase-Copper Hybrid Nanoflowers Embedded With Magnetic Nanoparticles as an Effective Antibacterial Agent," *International Journal of Biological Macromolecules* 155 (2020): 1520–1531, https://doi.org/10.1016/J.IJBIOMAC.2019.11.129.

86. K. Wang, C. Redshaw, X. Zhao, A. Wang, K. Chen, and Y. Qiao, "Silver Nanoparticle-Immobilized Schiff-Base Macrocycles as Nanozymes With Peroxidase Mimic Activity for Antibacterial Films," *ACS Applied Nano Materials* 7 (2024): 16274–16282, https://doi.org/10.1021/ACSANM.4C02258/SUPPL_FILE/AN4C02258_SI_001.PDF.

87. L. Tonoyan, D. Montagner, R. Friel, and V. O'Flaherty, "Antimicrobials Offered From Nature: Peroxidase-catalyzed Systems and Their Mimics," *Biochemical Pharmacology* 182 (2020): 114281, https://doi.org/10.1016/J.BCP.2020.114281.

88. S. Fu, H. Chen, H. Li, J. Duan, and H. Tan, "Self-activated Hydrogel Cascade Reactor Integrated With Glucose Oxidase and Silver Nanoparticle for Enhanced Treatment of Bacterial Infection," *International Journal of Biological Macromolecules* 277 (2024): 134081, https://doi.org/10.1016/J.IJBIOMAC.2024.134081.

89. M. Guo, Y. Liao, Y. Bao, et al., "Construction of GOx-Loaded Metal Organic Frameworks Antibacterial Composite Hydrogels for Skin Wound Healing," *International Journal of Biological Macromolecules* 295 (2025): 139655, https://doi.org/10.1016/J.IJBIOMAC.2025.139655.

90. Q. Wang, W. Qiu, M. Li, et al., "Mussel-Inspired Multifunctional Hydrogel Dressing With Hemostasis, Hypoglycemic, Photothermal Antibacterial Properties on Diabetic Wounds," *Biomaterials Science* 10 (2022): 4796–4814, https://doi.org/10.1039/D2BM00433J.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adma71881-sup-0001-SuppMat.pdf